Quantum Mechanics

Leonard Susskind

Core track, 2012 Winter Modern Physics Stanford
Lecture notes organized by LazyingArt LLC with Video2Book

Original lectures by Leonard Susskind. Lecture notes organized by LazyingArt LLC with Video2Book.

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Chapter 1

Modern Physics: Quantum Mechanics

This first lecture opens in an unusual way. Instead of beginning with a formal postulate list, Leonard Susskind begins by explaining the style of the course and then turns at once to a deeper claim: quantum mechanics is not classical mechanics with a little noise stirred in. To make that point, he unfolds the subject in the order the lecture itself demands. We begin with classical randomness as a foil, then the two-slit experiment, then reversibility and the cost of looking, then Heisenberg's microscope, and only after that do we take the first abstract step toward complex vector spaces. Where board frames curated through LazyingArt LLC and Video2Book help stabilize equations or layout, they are retained as supporting evidence; the transcript remains the primary source for the sequence and force of the argument.

1.1 Why This Course Starts With Logic, Not Machinery

Susskind begins with the audience. This is not freshman service physics, and it is not a standard undergraduate or graduate course. It is a continuing-education course, and that fact shapes the method. The audience wants the basic ideas quickly, but not cheaply. We are to use equations when necessary, and when we do use them, we are to use the simplest equations that will actually do the job. He even jokes that one might call it quantum mechanics for old people, including himself. The joke matters less than the pedagogical point behind it: we are not here for ornament, but neither are we here for slogans.

That brief preface also explains why the lecture opens with logic rather than machinery. Susskind reminds us that classical mechanics came first in the larger sequence of courses, but quantum mechanics is more fundamental. Classical physics is what we get when quantum mechanics may be approximated by classical logic in the appropriate regime. So instead of starting with Schrödinger's equation or operator formalism, the lecture starts by asking what classical logic gets wrong. That is the right place to begin, because the mathematics introduced later is meant to replace a structure that has already failed.

1.2 Classical Randomness Is Not Quantum Unpredictability

The first task is to separate two ideas that are too easily conflated. Quantum mechanics certainly contains unpredictability. But that does not mean that it is simply classical mechanics plus random

disturbances. To make the distinction sharp, Susskind invents a classical foil.

Imagine an otherwise Newtonian world in which a deity occasionally gives the moon tiny random kicks. Every tenth of a second the dice are thrown; on one outcome the moon is nudged one way, on another it is nudged the other way. This world would be statistically unpredictable, but the unpredictability would be of a familiar classical kind. It would enter by occasional deviations from deterministic law.

Now ask what such randomness would do. It would not merely blur our predictions. It would also jostle conserved quantities. If the kicks are random, then the energy of the system would, step by step, randomly increase or decrease. On average it might drift slowly or perhaps not at all, but exact conservation would be gone. The very form of the randomness would show itself in the energy bookkeeping.

Quantum mechanics does not behave that way. We prepare a system with a precise energy, let it evolve, and then measure again. The later result may be unpredictable in some respects, but the energy comes back exactly the same. No trace remains of a hidden hand knocking the system this way and that. So the lecture's first lesson is already in place: quantum unpredictability is highly special. It is not what one would get by sprinkling classical randomness into Newton's laws.

1.3 The Two-Slit Experiment and the Failure of Additive Probabilities

Susskind's first great example is the two-slit experiment, because it exhibits the weirdness of quantum mechanics in the most direct and least encumbered way. Let us follow the order of the lecture carefully.

We begin with a source of very light particles—photons, electrons, neutrons, anything for which quantum effects are appreciable. They are sent through the apparatus one at a time. This matters. The lecture insists on that sparsity precisely so that no later particle can plausibly be blamed for what happens to the earlier one.

First open only one small hole. A particle passes through and hits the screen in an isolated blip. Repeat the experiment many times and those individual blips build up a smooth profile: a bright central region fading away to the sides. We may call that one-hole distribution $P_1(y)$. Close that hole, open the second one, and repeat. We obtain another profile $P_2(y)$, displaced relative to the first.

At this stage classical logic seems irresistible. If both holes are open, then a particle reaching a point y on the screen must have come either through the first hole or through the second. Therefore the probabilities ought to add:

$$P_{12}(y) = P_1(y) + P_2(y). \tag{1.1}$$

This is not the quantum rule. It is the classical expectation that Susskind wants fully on the table before overturning it.

Now do the real experiment. The particles still come one at a time. The screen still records one blip at a time. But the built-up pattern is no longer the sum of the two one-hole profiles. Instead we get an interference pattern, and at some points the probability vanishes entirely:

$$P_{12}(y) = 0 \qquad \text{even though} \qquad P_1(y) \neq 0, \quad P_2(y) \neq 0. \tag{1.2}$$

This is the shock of destructive interference. With only the first hole open, particles arrive at that point. With only the second hole open, particles also arrive at that point. Open both holes, and no particle ever arrives there.

That is the point where Susskind widens the claim. One could try to invent some elaborate mechanism, some baroque memory hidden in the apparatus, but the phenomenon is too generic for that sort of one-off repair. The difficulty is not a local detail. The difficulty is that classical statistics and classical logic are built on the wrong underlying structure. The additive rule for alternatives has simply failed.

1.4 Reversibility, Information, and the Cost of Looking

Having displayed the failure of additive probabilities, the lecture shifts to a second diagnostic: reversibility. The move is deliberate. Susskind wants a different test, one that is not tied to slit interference, and for that he first backs up to very simple discrete systems.

Consider a two-state system, a coin with states H and T. A deterministic law of motion can take the form

$$H \mapsto H, \qquad T \mapsto T, \tag{1.3}$$

or it can alternate the states,

$$H \mapsto T, \qquad T \mapsto H. \tag{1.4}$$

Now move to a three-state system with heads, tails, and feet:

$$H \mapsto T, \qquad T \mapsto F, \qquad F \mapsto H. \tag{1.5}$$

The important point is not which law we choose. The important point is that in each of these examples the evolution is exact and information-preserving. If we let the system evolve for a long time and then reverse the arrows of the update rule, it retraces its history and returns to the starting state.

This gives Susskind an operational definition of reversibility. We do not need the detailed law; we need only the possibility of reversing it. Evolve for a time t, reverse the rule, evolve for the same time t again. If the system always returns to its starting point, the dynamics is deterministic in the sense he wants.

Now add a tiny bit of classical randomness. Suppose that with some very small probability the system does not follow the prescribed update and instead does something else—for example, it stays put when the rule says move on. Over short times the reversal test may still succeed. But over sufficiently long times a fluctuation will almost certainly occur, and then running backward does not return us uniquely to the initial state. Information has been lost. That is why Susskind says that classical randomness destroys the conservation of information.

He now carries the same test over into quantum mechanics. Here the lecture momentarily wavers in its wording, but the logic is clear: for this thought experiment we no longer need two slits. One slit is enough. Send one electron through the slit. Remove the screen. Let the electron pass through, but do not detect it. Then reverse the law of physics and run the system backward for the same length of time.

What happens? If nothing has looked at the electron, nothing has recorded its location, nothing has disturbed it in the intermediate stage, then the reversal is exact. The electron retraces the reverse trajectory and goes back into the gun. That is already a remarkable statement. The apparent randomness of the one-slit deflection is not the same thing as classical random compounding of errors.

Now alter only one thing. Put in a detector and record where the electron is during the intermediate stage. Then reverse the law and run backward. In that case the exact return fails. The uncertainty does not cancel; it compounds. The reversibility test, which classical gentle measurement would have left untouched, is destroyed by quantum measurement itself.

This is the deeper transition of the lecture. Classical physics allows an arbitrarily gentle observation. One can, in principle, look at a classical system without significantly altering its subsequent behavior. Quantum mechanics does not. Determining the state of the system is never a harmless afterthought.

Susskind then returns to the two-slit experiment and states the same idea in a different form. The interference pattern is present only if nothing in the apparatus or environment records which slit the electron went through. That recorder need not be a human observer. It could be a gas molecule, another electron, or any degree of freedom that remembers afterward which path was taken. Once such path information exists in the environment, the interference pattern disappears and the answer reverts to the classical one: the probabilities add.

1.4.1 Question & Answer

Question. If the particle is deflected by the slit or plate, does the plate not recoil? And if it recoils, does that not already reveal which path the particle took?

Answer. The recoil is real. The issue is whether it is readable. Susskind's answer is qualitative and deliberately anticipates the uncertainty principle before formally deriving it. A detector that is very well localized in position must have a broad momentum distribution. So although an electron gives the detector a small kick, that kick only shifts an already broad momentum distribution by a small amount. Unless the shift is comparable to the width of the distribution, one cannot tell afterward whether the kick occurred.

The lesson is precise enough for our purposes. Recoil does not automatically provide which-path information. It provides readable information only if the shift in detector momentum is large compared with the detector's own intrinsic momentum uncertainty.

1.5 Heisenberg's Microscope and the Momentum of a Photon

At this point Susskind turns explicitly to the uncertainty principle. But again he does not present it first as a formal theorem. He stages it as a demand for physical explanation. Heisenberg may say that position and momentum cannot be simultaneously determined, but Bohr's challenge is: do not merely tell me that x and p fail to commute; show me the physics that forces this obstruction.

The physical setting is measurement by light. To locate a particle under a microscope, we scatter a photon from it, collect the scattered light, and form an image. So the next question is simple: what did Einstein, de Broglie, Bohr, and Heisenberg know about photons that makes this measurement problematic?

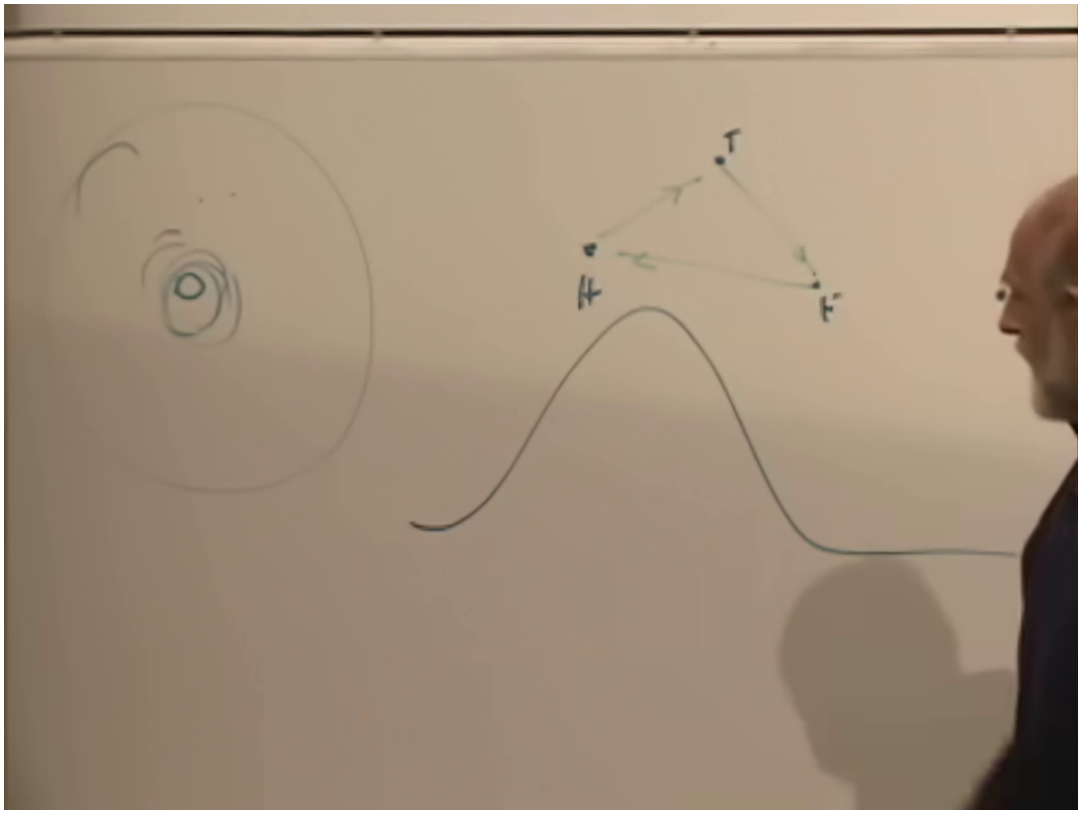

Figure 1.1: Board layout for the recoil argument. The broad central hump is the detector momentum distribution introduced in the answer; the large oval at left and the small triangle-like sketch at upper right are residual board context and should not be overread.

The board snapshot is valuable here because it preserves the cluster of relations as Susskind arranged them.

The first relation is Einstein's photon-energy formula:

$$E = hf = \hbar\omega, \tag{1.6}$$

with

$$\omega = 2\pi f, \qquad \hbar = \frac{h}{2\pi}. \tag{1.7}$$

Susskind then recalls that light carries momentum as well as energy. For ordinary nonrelativistic particles, one has

$$E = \frac{p^2}{2m} = \frac{1}{2}pv, \tag{1.8}$$

so energy and momentum already have the right dimensions to be related by a speed. For light, the relativistic relation is simpler:

$$E = cp, \qquad \text{hence} \qquad p = \frac{E}{c}. \tag{1.9}$$

Now combine this with the photon-energy formula:

$$p = \frac{hf}{c}. \tag{1.10}$$

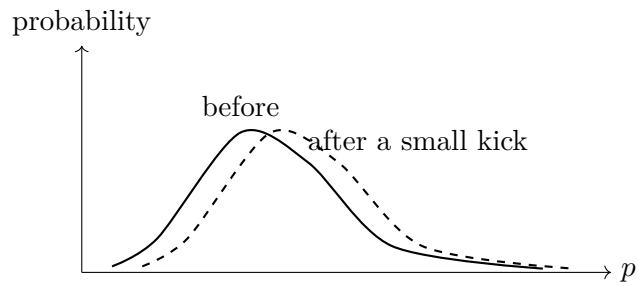

Figure 1.2: Transcript-derived redraw of the detector momentum distribution. The lecture's point is that a small recoil may shift the distribution without making that shift experimentally readable.

To eliminate the frequency we use the kinematics of a wave. If one cycle takes time T, then

$$T = \frac{1}{f}. \tag{1.11}$$

If the wavelength is λ, then in one period the wave advances by one wavelength, so

$$c = \frac{\lambda}{T} = \lambda f, \qquad \text{or equivalently} \qquad f = \frac{c}{\lambda}. \tag{1.12}$$

Substituting into the momentum relation gives the de Broglie formula:

$$p = \frac{hf}{c} = \frac{h}{c}\frac{c}{\lambda} = \frac{h}{\lambda}. \tag{1.13}$$

This is the decisive step: momentum and wavelength are inverse to one another.

The lecture then pauses and leans on the physical meaning of that inverse relation. If we want to form an image that is sharp on a scale Δx, then the wavelength used to form the image must be at most of that order:

$$\lambda \lesssim \Delta x. \tag{1.14}$$

That is not a fancy theorem here; it is ordinary imaging logic. A wavelength much larger than the desired detail necessarily produces a fuzzy image on that larger scale.

But now the trouble appears. If λ must be small, then the photon momentum must be large:

$$p_\gamma = \frac{h}{\lambda}. \tag{1.15}$$

So the very photon that makes a sharp image is also a high-momentum projectile. When it scatters from the particle whose position we are trying to determine, it gives that particle an uncontrollable momentum kick. The lecture does not derive a sharp inequality such as $\Delta x\,\Delta p \geq \hbar/2$, and we should not import one prematurely. What it does justify is the order-of-magnitude statement

$$\Delta p_{\text{kick}} \sim \frac{h}{\lambda}. \tag{1.16}$$

Thus a precise position measurement necessarily produces a large uncertainty in the post-measurement momentum.

Susskind then points out why classical physics seemed to evade this difficulty. In classical physics, light is not quantized. One imagines that one can keep the wavelength fixed while making the

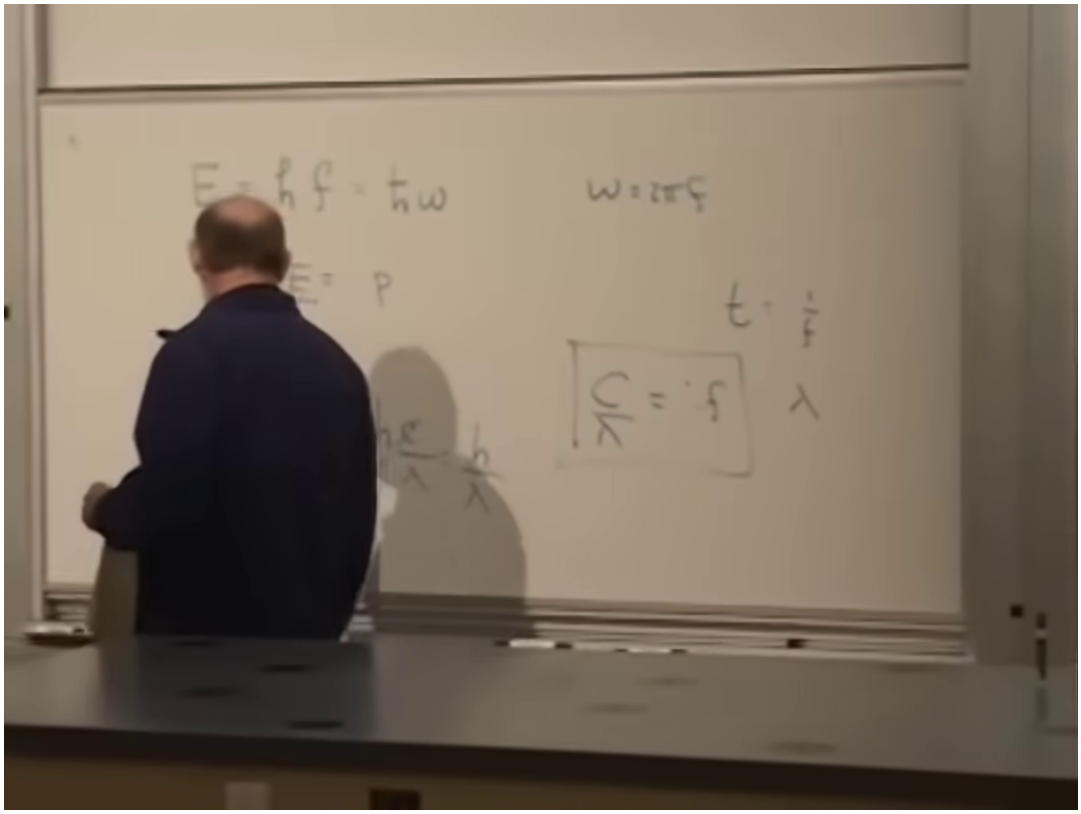

Figure 1.3: Board layout for the photon-energy and wave-kinematics relations leading to the inverse momentum-wavelength law.

intensity arbitrarily small, thereby preserving resolution without delivering a meaningful kick. Quantum mechanics blocks that maneuver. At fixed wavelength there is a minimum packet, namely one photon, and one photon already carries momentum h/λ. There is no infinitely gentle probe of a given short wavelength.

At precisely this stage of the lecture, Susskind remarks that he had erased the key formula and rewrites it. The chalk line itself is partly hidden in the frame, so the reconstruction must be guided by the transcript and the surrounding visible equations.

The physical conclusion is now inescapable. There is no perfectly gentle sharp measurement of position in quantum mechanics. If the measurement is sharp, the momentum is badly disturbed. If the measurement is gentle, the position information is necessarily coarse.

1.5.1 Question & Answer

Question. If a sharp position measurement ruins momentum, how can we measure a particle's velocity at all?

Answer. We measure it gently, and we take our time. The lecture's answer is operational. Measure the position twice, at two different times, and infer the velocity from the displacement divided by the elapsed time. But each position measurement must be gentle, so each must use a long-wavelength

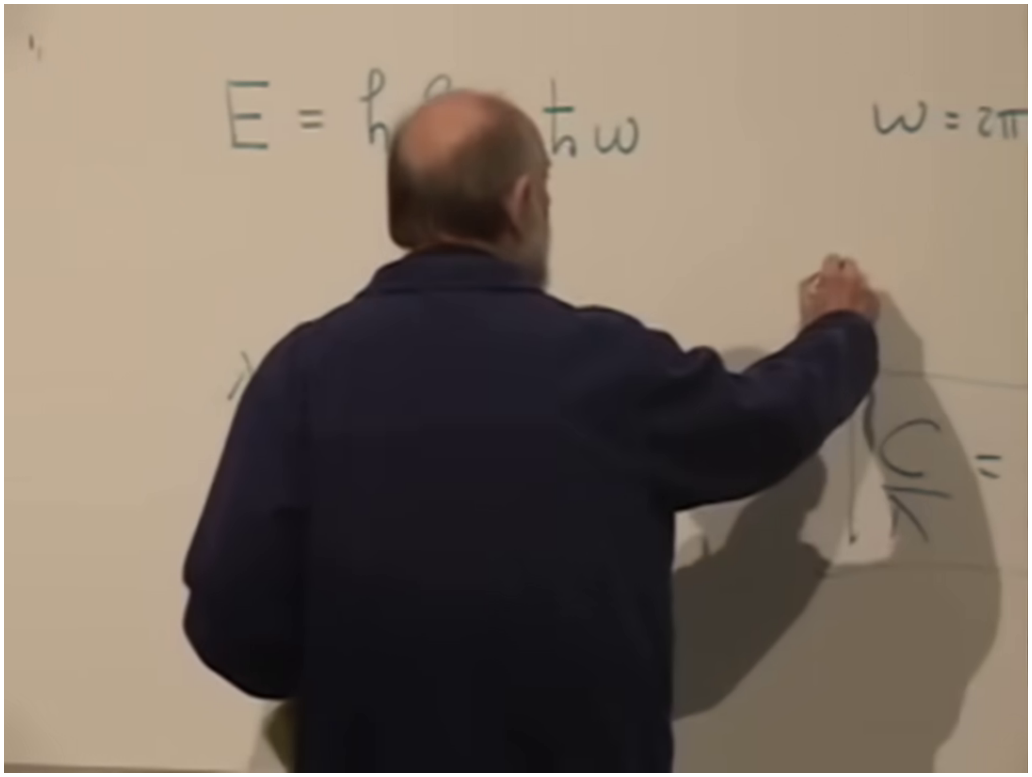

Figure 1.4: Susskind rewriting the erased momentum-wavelength relation. The exact chalk line is partly occluded, but the transcript makes clear that the restored equation is $p = h/\lambda$.

photon.

A long-wavelength photon localizes the particle only coarsely:

$$x_1 = x \pm O(\lambda), \tag{1.17}$$

$$x_2 = x + vt \pm O(\lambda). \tag{1.18}$$

Subtracting the two measurements gives the traveled distance,

$$d = x_2 - x_1 = vt \pm O(\lambda). \tag{1.19}$$

Therefore the inferred velocity has uncertainty

$$\Delta v \sim \frac{\lambda}{t}. \tag{1.20}$$

If we make t very large, then Δv can be made very small even when λ is large. So velocity may indeed be measured accurately. But the price is that the position was never known to better than order λ. This is exactly the complementary tradeoff Susskind wants us to keep in mind. We can know velocity well, but only by giving up precise knowledge of position; or we can localize sharply, but only by badly disturbing the momentum.

1.6 From Classical States to Quantum Vector Spaces

Only after all of these physical examples have done their work does the lecture make its final turn. Susskind says, in effect: now we are ready to ask what is really new in the mathematical language.

In classical mechanics, the state of a system is a point in a set. That already tells us something about the operations available. We can ask whether two sets intersect or unite, but it makes no sense to multiply a point by a number. If the only states are heads and tails, then $3 \times$ heads and $(2 + 4i) \times$ heads are meaningless expressions.

Quantum mechanics will replace this set-like notion of state by a vector-space notion. That should feel radical, and Susskind wants it to feel radical. He even says that at this point it should sound like mumbo jumbo. But the lecture is careful: before asking why physical states should be vectors, it first explains what a vector space is.

Definition 1.1. A complex vector space is a collection of objects for which two operations are defined: any two objects can be added, and any object can be multiplied by an arbitrary complex number, with the result again belonging to the same collection.

Once those operations exist, any linear combination is again an allowed object:

$$\alpha A + \beta B \in V \qquad \text{whenever} \qquad A, B \in V,\ \alpha, \beta \in \mathbb{C}. \tag{1.21}$$

That is the basic algebra the lecture wants us to absorb.

Ordinary geometric pointers provide the familiar analogy. They form a vector space over the real numbers: they may be added, and they may be multiplied by real scalars. But quantum mechanics wants something more general, namely vector spaces over the complex numbers.

The first example Susskind gives is the space of complex-valued functions of a real variable x. Here x itself is an ordinary real variable; what is complex is the value of the function:

$$\psi(x) \in \mathbb{C}. \tag{1.22}$$

If $\psi(x)$ and $\phi(x)$ are such functions, and if α is a complex number, then

$$(\alpha\psi)(x) = \alpha\,\psi(x), \tag{1.23}$$

$$(\psi + \phi)(x) = \psi(x) + \phi(x). \tag{1.24}$$

Both operations produce new complex-valued functions of x. Therefore complex functions of one real variable form a complex vector space. This is the example that will later become central in quantum mechanics.

The lecture also makes an important contrast. Real-valued functions form a vector space over the real numbers, but not over the complex numbers. The reason is immediate: multiplying a real-valued function by i does not keep it real-valued.

The second example is the fixed-length column vector. If we take a column of four arbitrary complex numbers,

$$A = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix}, \qquad B = \begin{pmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \end{pmatrix}, \tag{1.25}$$

then addition is defined entry by entry:

$$A + B = \begin{pmatrix} A_1 + B_1 \\ A_2 + B_2 \\ A_3 + B_3 \\ A_4 + B_4 \end{pmatrix}, \tag{1.26}$$

and multiplication by a complex scalar is equally simple:

$$\alpha A = \begin{pmatrix} \alpha A_1 \\ \alpha A_2 \\ \alpha A_3 \\ \alpha A_4 \end{pmatrix}. \tag{1.27}$$

For a fixed length, these columns form a vector space. But here the lecture is careful about a subtle point: the collection of all columns of all lengths is not a single vector space under these rules, because no addition law has been given between, say, a two-component vector and a three-component one.

Finally, Susskind steps back to the complex numbers themselves. They already form a one-dimensional complex vector space. That makes complex conjugation the next natural operation to inspect. Write

$$z = x + iy, \qquad z^* = x - iy. \tag{1.28}$$

Geometrically, z^* is the reflection of z across the real axis. Algebraically,

$$z^* z = (x + iy)(x - iy) = x^2 + y^2. \tag{1.29}$$

So the product of a number with its conjugate gives the square of its magnitude. Susskind treats this as a basic operation, not a decorative trick, because it is one of the first examples of a mapping that will later be generalized from numbers to vectors.

The closing note of the lecture is important. We are not yet being told in what precise sense physical states are vectors. That comes later. For now we are learning the abstract operations themselves. And that is exactly right: if classical logic is built on sets, then quantum logic will need a new arena, and the lecture ends by introducing the first pieces of that arena.

1.7 Summary

The lecture begins by rejecting the easy idea that quantum mechanics is just classical mechanics with random noise added. Classical random kicks would spoil exact conservation laws and destroy reversibility. Quantum mechanics does neither in that simple way. The two-slit experiment then shows that probabilities do not merely add: with both slits open, there are points where the probability vanishes even though each one-slit experiment separately gives nonzero arrivals there.

The lecture then turns the notion of reversibility into an operational test of determinism. Classical randomness destroys that test; quantum evolution preserves exact reversibility only so long as nothing records intermediate information. That is the deeper lesson behind both the one-slit reversal experiment and the two-slit which-path story: in quantum mechanics, measurement is part of the dynamics.

Heisenberg's microscope then gives the physical mechanism. To resolve position sharply we need short wavelength; short wavelength means large photon momentum; the very photon that makes the image gives the particle an uncontrollable kick. Velocity can still be measured well, but only by very gentle long-wavelength position measurements spread over a long time, and therefore only at the cost of poor position information.

Only after these physical examples have dismantled the classical picture does Susskind make the mathematical pivot. Classical states are points in a set. Quantum states will live in a complex

vector space. Functions, column vectors, and complex numbers themselves are the first examples. That is where the lecture stops: not yet with the full machinery of operators and eigenvalues, but with the clear sense that some new logic is required, and with the first abstract language in which that logic can be expressed.

Chapter 2

Vector Spaces, Superposition, and the First Quantum Postulates

These notes follow Leonard Susskind's second quantum mechanics lecture as a guided argument rather than as a list of facts. The lecture begins by removing a conceptual obstruction left over from the discussion of the uncertainty principle, then turns to the formal language of complex vector spaces, bras and kets, inner products, bases, and finally the first quantum-mechanical postulates in the simple language of coins and dice. The selected lecture frames are kept where the board layout or the chalk geometry matters, but the transcript remains the primary source of order and emphasis. In that spirit we keep explicit credit to Leonard Susskind, together with the lecture-frame curation work of LazyingArt LLC and Video2Book.

2.1 Uncertainty Principle Prelude

The lecture does not begin with vector spaces. It begins by clearing away an objection. Suppose we measure the position of a particle only very poorly, to within a region of size δ, and then wait a very long time T before measuring its position again, again only to within a region of size δ. If the centers of the two fuzzy regions are separated by a distance L, then one seems to know the average velocity very accurately:

$$v_{\text{avg}} \sim \frac{L}{T}, \tag{2.1}$$

$$\Delta v_{\text{avg}} \sim \frac{\delta}{T}. \tag{2.2}$$

Since T may be made arbitrarily large, the uncertainty δ/T may be made arbitrarily small. At first sight this looks like a way of knowing position only to accuracy δ, while knowing velocity essentially exactly.

That, however, is not the quantity that the uncertainty principle is about. What we wanted was the position and momentum of the same state of the particle, or at least the momentum just before the position measurement. The two-measurement trick does not return that momentum.

2.1.1 Question & Answer

Can two imprecise position measurements defeat the uncertainty principle?

No. The first position measurement already disturbs the momentum. To localize a particle to scale δ, one must probe it with radiation of wavelength of order δ or smaller, and this produces a momentum kick of order

$$\Delta p \sim \frac{\hbar}{\delta}. \tag{2.3}$$

So the later motion reveals the velocity after the first measurement, not the original velocity of the undisturbed state. The uncertainty principle is not being evaded; rather, the wrong velocity is being inferred.

The lecture also presses the point of repeatability. One may say, "I think I measured the velocity," but if one tries to check that velocity one changes it again by an amount of order $\hbar/\delta$. In classical physics one imagines measuring position arbitrarily gently and leaving the velocity alone. In quantum mechanics, not so.

With that conceptual obstacle removed, the lecture pivots very explicitly: now we are ready to begin.

2.2 Complex Numbers as the Simplest Complex Vector Space

The next step is intentionally modest. Susskind says he is not trying to teach complex numbers; he wants only a very quick review, just for notational reasons. The subject of linear vector spaces may be dry, he says, but the good news is that it is also easy. So we begin with the easiest complex vector space of all: the complex numbers themselves.

A complex number is

$$z = x + iy, \qquad x, y \in \mathbb{R}, \tag{2.4}$$

and its complex conjugate is

$$z^* = x - iy. \tag{2.5}$$

Geometrically, z^* is the reflection of z across the real axis in the complex plane. The lecture briefly reminds us not to confuse that plane with the vector space we ultimately care about. The point of the example is structural: complex numbers can be added, and they can be multiplied by complex numbers, so they already form a one-dimensional complex vector space.

At this stage the bra-ket notation is introduced in embryo. In the one-dimensional case one may think of z as a ket and z^* as the corresponding bra. This is not yet deep; it is just a simple rehearsal for what will happen in more complicated vector spaces.

The lecture then isolates one algebraic fact that will return later in a more abstract form. If

$$z_1 = x_1 + iy_1, \qquad z_2 = x_2 + iy_2, \tag{2.6}$$

then

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2). \tag{2.7}$$

Complex conjugation flips only the sign of the imaginary part, and therefore

$$(z_1 z_2)^* = z_1^* z_2^*. \tag{2.8}$$

A related identity, written in the notation that foreshadows bras and kets, is

$$(z_1^* z_2)^* = z_2^* z_1. \tag{2.9}$$

For ordinary numbers the order is harmless, since the factors commute. But the lecture lingers on this order-swapping because it prepares the listener for the later statement that interchanging bra and ket corresponds to complex conjugation.

Only after this warm-up does the lecture state the abstract idea. A complex vector space is made of objects that can be added and multiplied by arbitrary complex numbers, and that stay within the space under those operations. In particular, if A and B are vectors and $\alpha, \beta \in \mathbb{C}$, then

$$\alpha A + \beta B \tag{2.10}$$

is again a vector of the same space.

That definition is deliberately spare. Vectors are not defined as arrows in physical space. They are defined by the operations we are allowed to perform on them. Ordinary arrows form only a real vector space, because multiplication by a general complex number is not defined for them. The real numbers form a real vector space, but not a complex one. Even the positive real numbers fail to form a real vector space, because multiplication by a negative number takes us out of the set. This operational point of view is the real beginning of the mathematics.

2.3 Complex Vector Spaces: Examples, Dimension, and Abstract Notation

Once the definition is on the table, the lecture immediately asks for examples. The first important example is the space of complex-valued functions of a real variable:

$$\psi(x) = \psi_{\mathrm{Re}}(x) + i\,\psi_{\mathrm{Im}}(x). \tag{2.11}$$

Here x is real, but the value of $\psi(x)$ is complex. If we multiply $\psi(x)$ by a complex number, we get another complex-valued function. If we add two such functions, we get another. In this mathematical sense, functions are vectors.

The second example is the column vector. A column vector is really a function of a discrete index. In the four-component case we may write

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, \qquad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}, \tag{2.12}$$

and define the vector-space operations component by component:

$$|a\rangle + |b\rangle = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \\ a_4 + b_4 \end{pmatrix}, \tag{2.13}$$

$$\alpha|a\rangle = \begin{pmatrix} \alpha a_1 \\ \alpha a_2 \\ \alpha a_3 \\ \alpha a_4 \end{pmatrix}. \tag{2.14}$$

Thus column vectors are simply the discrete analog of complex functions. The lecture is careful here: if one is used to thinking of vectors only as arrows, one may not be used to thinking of functions as vectors. But that unfamiliarity is psychological, not mathematical.

Having supplied examples, the lecture turns to dimension. The dimension of a vector space is defined as the minimum number of vectors needed so that every vector in the space can be written as a linear combination of them. On a line, one nonzero vector suffices. On the blackboard plane, two non-collinear vectors suffice. This is the point at which the lecture draws the spanning picture.

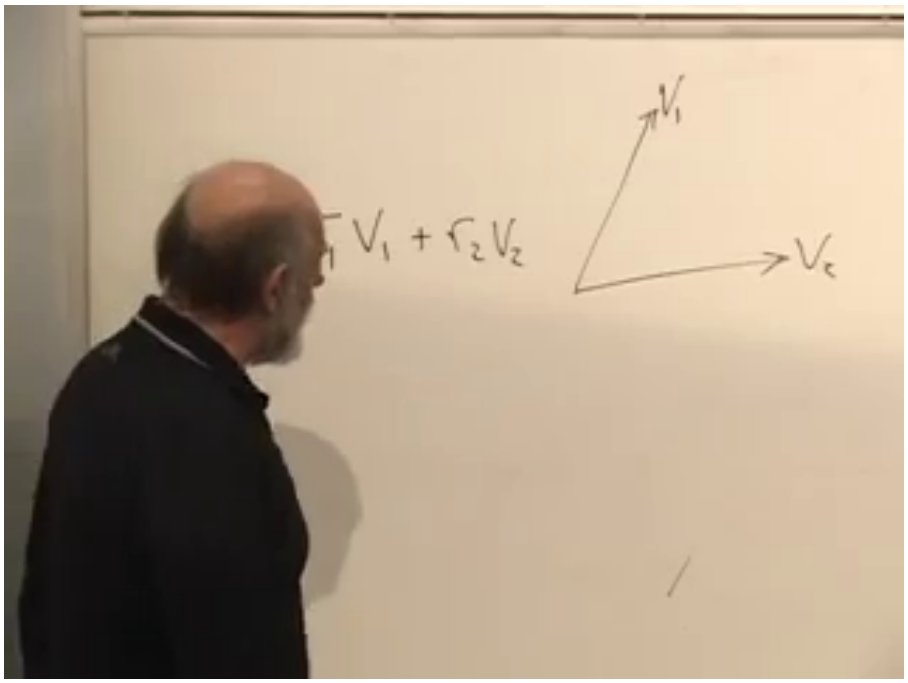

Figure 2.1: Linear combination from two spanning vectors. The chalk labels look closer to uppercase V_1, V_2, while the notes normalize the mathematics to lowercase v_1, v_2.

The chalk expression at the left of the board is partially obscured, but the clean mathematical statement is straightforward: if v_1 and v_2 are not on the same line, then every vector in the plane can be written as

$$r_1 v_1 + r_2 v_2. \tag{2.15}$$

A third vector v_3 drawn in the wedge between them would be redundant, because it can already be expressed in terms of the first two:

$$v_3 = r_1 v_1 + r_2 v_2. \tag{2.16}$$

The selected frame does not actually show v_3; that vector comes from the spoken argument and is drawn only in the cleaned reconstruction below.

From here the lecture runs through a sequence of dimensions. A four-component column space is four-dimensional. A one-component column space is just the one-dimensional complex vector space of ordinary complex numbers. Smooth functions of a real variable form an infinite-dimensional vector space, because no finite set of functions can be combined to generate every smooth function.

Only after that progression does the lecture shift notation. The point is to detach the reader from any one representation and move to the abstract vector itself.

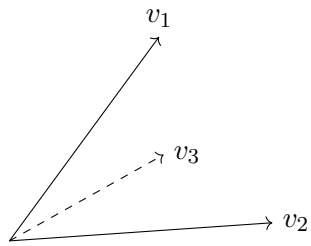

Figure 2.2: Clean reconstruction of the spanning argument. The dashed vector v_3 is reconstructed from the lecture discussion and is not visibly completed in the screenshot.

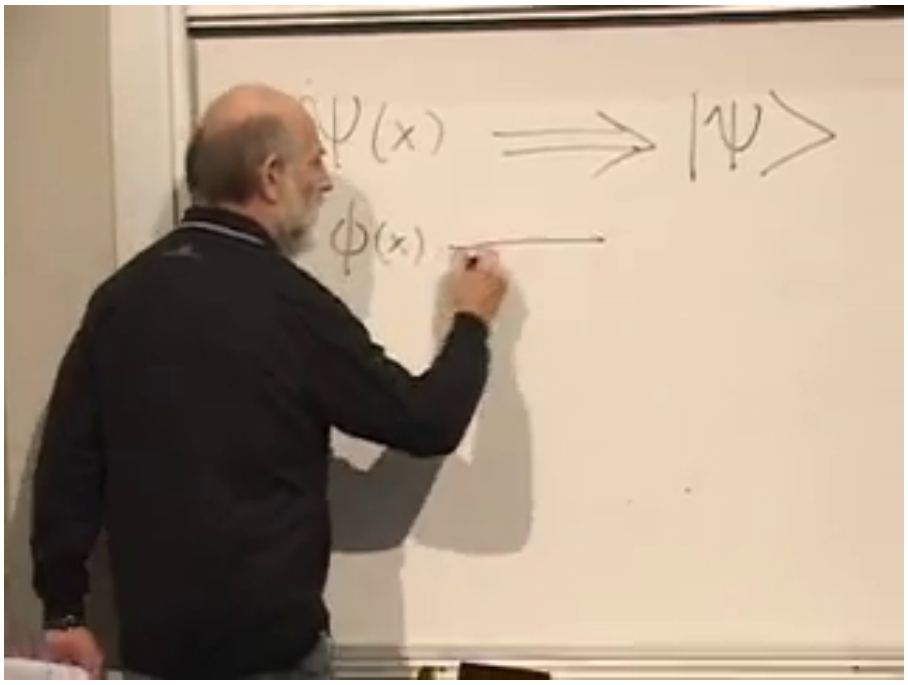

Figure 2.3: Mapping $\psi(x)$ to ket notation. The upper relation is fully visible on the board; the lower line is only partially visible and is completed cautiously in the text.

The board shows the completed line

$$\psi(x) \Rightarrow |\psi\rangle \tag{2.17}$$

and begins a second analogous line. In the text we complete that second line cautiously as

$$\phi(x) \mapsto |\phi\rangle. \tag{2.18}$$

The arrow here is not a strict equality sign. It is a change of viewpoint. We are now to think of the function as a vector in an abstract complex vector space. In the same way, a column of entries may be thought of as the abstract vector determined by those entries. This is the point at which the lecture's working notation settles into place.

2.4 Dual Vectors and Inner Products

Once abstract vectors are in place, the lecture introduces the dual vector space. For the purposes of this course, the dual space is the complex-conjugate partner of the original ket space. For every ket $|A\rangle$ there is a corresponding bra $\langle A|$. Mathematicians can define duals more generally, but the physical case under discussion is the complex-conjugate one.

In the case of functions, the bra corresponding to $|\psi\rangle$ is represented by the complex-conjugate function. In the case of column vectors, the corresponding bra is a row vector whose entries are the complex conjugates of the entries of the column. The lecture emphasizes that some of this is notation, but notation chosen to keep track of which objects live in the ket space and which in the bra space.

The first nontrivial rule concerns scalar multiplication. If $\langle A|$ corresponds to $|A\rangle$, then the bra associated with $\alpha|A\rangle$ is

$$\alpha^*\langle A|, \tag{2.19}$$

not $\alpha\langle A|$. This is just the familiar rule that complex conjugation of a product conjugates both factors. The lecture even pauses over a student's question about whether the α^* has to be written on the right; for ordinary numbers it does not matter, though later it will matter once operators are introduced.

The next construction is the inner product. Susskind introduces it by analogy with the ordinary dot product of real vectors. For ordinary vectors,

$$a \cdot b = a_x b_x + a_y b_y + a_z b_z + \cdots . \tag{2.20}$$

The general inner product $\langle A|B\rangle$ is the complex analogue of this idea, but now one must decide which vector is the bra and which is the ket. Once that choice is made, the inner product is required to satisfy the following properties:

$$\langle A|(\beta B)\rangle = \beta\langle A|B\rangle, \tag{2.21}$$

$$\langle A|(B+C)\rangle = \langle A|B\rangle + \langle A|C\rangle, \tag{2.22}$$

$$\langle A|B\rangle = \langle B|A\rangle^*, \tag{2.23}$$

$$\langle A|A\rangle > 0 \qquad \text{for } |A\rangle \neq 0. \tag{2.24}$$

From the third relation we learn immediately that $\langle A|A\rangle$ is real, since it equals its own complex conjugate. But the lecture insists on something subtler: positivity does *not* follow from the other properties. It is an additional part of the definition. Susskind even mentions relativity as an example where a bilinear form on a real vector space need not be positive.

The scalar model for positivity is the elementary identity

$$z^* z = (x - iy)(x + iy) = x^2 + y^2, \tag{2.25}$$

which is real and nonnegative. That is the pattern we carry into vector spaces.

The lecture then gives the concrete inner products we will actually use. For complex-valued functions of a real variable,

$$\langle\phi|\psi\rangle = \int dx\, \phi^*(x)\psi(x). \tag{2.26}$$

The lecture allows that functions of several real variables would simply require several integrations, but keeps the arguments real and the example simple.

For column vectors, if

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \qquad \langle b| = \begin{pmatrix} b_1^* & b_2^* & b_3^* \end{pmatrix}, \tag{2.27}$$

then

$$\langle b|a\rangle = b_1^* a_1 + b_2^* a_2 + b_3^* a_3 = \sum_i b_i^* a_i. \tag{2.28}$$

The lecture briefly corrects itself on the blackboard here, because one must remember that the row entries are the conjugates. In the final notes it is best simply to present the corrected form.

2.5 Orthonormal Bases and Expansion Coefficients

After the break the lecture becomes constructive again. We now introduce basis vectors, and again the order of ideas matters. First come unit length, then orthogonality, then orthonormality, and only then the practical use of the basis.

Let $\{|b_i\rangle\}_{i=1}^d$ be a collection of vectors in a d-dimensional vector space. They are normalized if

$$\langle b_i|b_i\rangle = 1 \qquad \text{for each } i, \tag{2.29}$$

with no sum on i. They are orthogonal if

$$\langle b_i|b_j\rangle = 0 \qquad \text{for } i \neq j. \tag{2.30}$$

Together these two conditions are summarized by

$$\langle b_i|b_j\rangle = \delta_{ij}, \tag{2.31}$$

where the Kronecker delta is defined by

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{2.32}$$

A set of vectors satisfying this relation is orthonormal, and if there are d of them in a d-dimensional space, they form a basis.

The lecture illustrates this with the standard basis for a four-component column space. In cleaned notation we may write

$$|b_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |b_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |b_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |b_4\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{2.33}$$

The corresponding bras are the conjugate row vectors. These vectors are normalized because each has only one nonzero entry, and they are orthogonal because two distinct basis vectors never overlap in the same slot. The lecture corrects a bra-versus-ket slip while discussing this example; here we simply present the corrected basis cleanly.

Now comes the practical payoff. Any vector $|v\rangle$ in a d-dimensional space can be expanded as

$$|v\rangle = \sum_{i=1}^{d} v_i |b_i\rangle. \tag{2.34}$$

The coefficients v_i are determined by taking inner products with the basis vectors. If we multiply on the left by $\langle b_j|$, we obtain

$$\langle b_j|v\rangle = \sum_{i=1}^{d} v_i \langle b_j|b_i\rangle \tag{2.35}$$

$$= \sum_{i=1}^{d} v_i \delta_{ji} \tag{2.36}$$

$$= v_j. \tag{2.37}$$

So the expansion coefficients are

$$v_j = \langle b_j|v\rangle. \tag{2.38}$$

This is the sequence the lecture wants us to see: first expand, then take the inner product, then use orthonormality, then discover the coefficients. Substituting back gives the compact formula that will recur throughout the course:

$$|v\rangle = \sum_{i=1}^{d} |b_i\rangle\langle b_i|v\rangle. \tag{2.39}$$

In the lecture the local blackboard derivation is a little messy; in the notes it is best presented in this cleaned form while preserving the same logical order.

2.6 Quantum States as Vectors: Coin, Die, Superposition, and Normalization

Only after the mathematics has been assembled does the lecture turn to physical meaning. The contrast with classical mechanics is drawn sharply. In classical mechanics, the space of states is a set. One speaks of unions and intersections, of AND and OR. In quantum mechanics, states are vectors, and the algebra is altogether different.

The lecture begins with the coin. Classically there are two states, heads and tails. One may imagine labeling them by symbols, but there would be no point in taking linear combinations of those labels. In quantum mechanics, by contrast, we assign to the two distinguishable configurations two orthogonal basis vectors, which we denote by $|H\rangle$ and $|T\rangle$. These are not lengths $+1$ and -1; they are simply names for two basis states. Their mathematical properties are

$$\langle H|T\rangle = 0, \qquad \langle H|H\rangle = \langle T|T\rangle = 1. \tag{2.40}$$

The orthogonality is not arbitrary. It is a postulate with physical meaning: states that can be distinguished by a single clean measurement are represented by orthogonal vectors. For the coin, the clean measurement is simply to look and ask whether the outcome is heads or tails. The same principle later applies to spin up and spin down, which is the real quantum-mechanical example lurking behind the coin.

Once the basis states are in place, the new quantum feature appears. The expression

$$|cc\rangle = \alpha_H|H\rangle + \alpha_T|T\rangle \tag{2.41}$$

is meaningful. Susskind calls this the state of a "confused coin." In classical physics the linear combination would mean nothing at all. In quantum mechanics it is a genuine state, different from both $|H\rangle$ and $|T\rangle$. The same idea applies to the die:

$$|cd\rangle = \sum_{i=1}^{6} \alpha_i|i\rangle, \tag{2.42}$$

where $|1\rangle, \ldots, |6\rangle$ are six orthonormal basis states corresponding to the six distinguishable outcomes of a die. Again, the labels $1, \ldots, 6$ are names, not vector lengths.

2.6.1 Question & Answer

How do the amplitudes become probabilities?

At this stage the lecture treats this as a postulate, not as something derived. If the confused coin is in the state

$$|cc\rangle = \alpha_H|H\rangle + \alpha_T|T\rangle, \tag{2.43}$$

then the probability of finding heads is

$$p_H = \alpha_H^*\alpha_H = |\alpha_H|^2, \tag{2.44}$$

and the probability of finding tails is

$$p_T = \alpha_T^*\alpha_T = |\alpha_T|^2. \tag{2.45}$$

This is the Born rule in its simplest discrete form.

The lecture is very careful here. This is not said to be the whole meaning of the amplitudes α_H and α_T. There is more in them than probabilities. But one feature of their meaning is already fixed: the squared moduli are the probabilities for measurement outcomes in the chosen orthonormal basis.

Operationally, these probabilities are understood from large ensembles of identically prepared systems. One prepares many copies of the same state $|cc\rangle$, performs the same measurement on each copy, and reads probabilities from the frequencies of outcomes. That is the level at which the postulate meets experiment.

Normalization now becomes unavoidable:

$$|\alpha_H|^2 + |\alpha_T|^2 = 1. \tag{2.46}$$

In bra-ket language this is simply

$$\langle cc|cc\rangle = 1. \tag{2.47}$$

Likewise, for the confused die,

$$\sum_{i=1}^{6} |\alpha_i|^2 = 1, \qquad \langle cd|cd\rangle = 1. \tag{2.48}$$

Thus the lecture arrives at another postulate: physically admissible states are normalized.

A useful explicit example is the equal-probability state. If the confused coin gives heads and tails with equal probability, then

$$|\alpha_H| = |\alpha_T| = \frac{1}{\sqrt{2}}. \tag{2.49}$$

But the amplitudes are complex, so the general equal-probability form is

$$\alpha_H = \frac{e^{i\theta_H}}{\sqrt{2}}, \qquad \alpha_T = \frac{e^{i\theta_T}}{\sqrt{2}}. \tag{2.50}$$

Similarly, an equal-probability die state may be written as

$$\alpha_i = \frac{e^{i\theta_i}}{\sqrt{6}}. \tag{2.51}$$

The lecture stresses exactly this point: equal probability does not mean the amplitudes are literally $1/\sqrt{2}$ or $1/\sqrt{6}$; it means they have those moduli, up to unit complex phases.

By the end of this section the first postulates have been stated in the language of simple discrete systems:

1. Clearly distinguishable configurations correspond to orthogonal vectors.
2. Physical states are normalized.
3. In a basis of distinguishable states, probabilities are given by squared moduli of amplitudes.

The deeper meaning of superposition is not yet exhausted, but the lecture has now established its first operational consequences.

2.7 Operators as the Next Mathematical Step

There is one more turn before the lecture ends. Susskind says he would like to get to operators, and he does, but only as an opening move. The subject is introduced first through ordinary blackboard vectors, before any full physical interpretation is attached to it.

The sketch is intentionally elementary: a vertical guide and two ordinary vectors. The question is what kinds of operations one can do on all vectors in such a space. One may rotate every vector by the same angle, double every vector while keeping its direction fixed, or reflect every vector about an axis. These are genuine operations on the vector space.

Not every operation is linear. The lecture gives a deliberately simple counterexample: replace every vector of length 2 by one of length 4, every vector of length 3 by one of length 9, and so on. That is certainly an operation, but it is not linear.

A linear operator, denoted with a hat, is characterized by two properties:

$$\hat{L}(\alpha a) = \alpha\, \hat{L}(a), \tag{2.52}$$

$$\hat{L}(a + b) = \hat{L}(a) + \hat{L}(b). \tag{2.53}$$

The lecture checks these properties on the simplest geometric example. If R denotes rotation by a fixed angle, then

$$R(2a) = 2R(a), \tag{2.54}$$

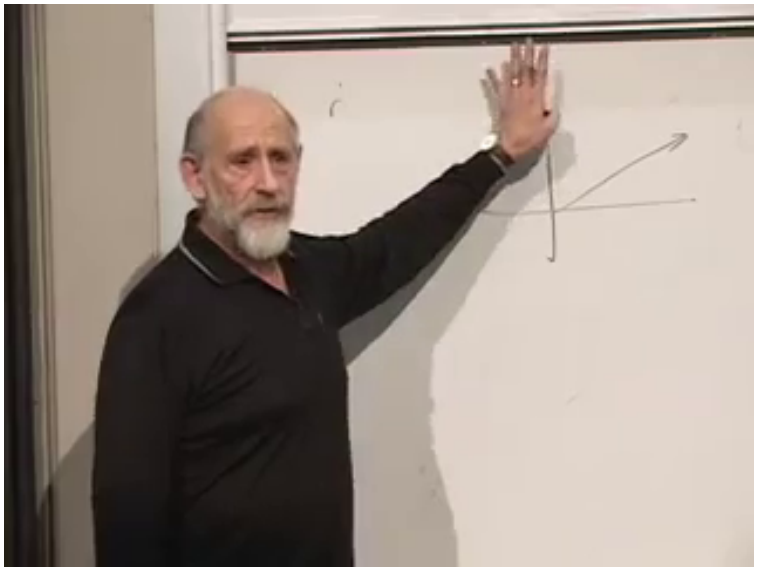

Figure 2.4: Ordinary vectors drawn on the blackboard. The figure is diagrammatic rather than symbolic, and serves as evidence for the geometric setup used to introduce operators.

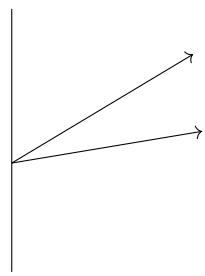

Figure 2.5: Clean redraw of the ordinary-vector sketch used in the lecture's first discussion of operations on vectors.

and if $a+b$ is drawn as the diagonal of the usual parallelogram, then rotating the whole parallelogram shows that

$$R(a+b) = R(a) + R(b). \tag{2.55}$$

So fixed-angle rotation is linear. Uniform doubling is linear. Reflection is linear. Squaring the length is not.

This final section is explicitly transitional. The lecture reaches just far enough to state the physical slogan: states are represented by vectors, but measurable quantities are represented by linear operators. In that sense observables such as the "headsness" or "tailsness" of the coin belong on the operator side of the formalism. The actual identification of observables with operators is postponed to the next lecture.

2.8 Summary

This lecture unfolds in two layers. First it clears away a conceptual misunderstanding about the uncertainty principle. Then it builds the mathematical language in which the first quantum

postulates can be stated.

We began by seeing why two poor position measurements far apart in time do not reveal the original momentum of a particle: the first position measurement itself changes the momentum by an amount of order $\hbar/\delta$. From there the lecture reset and moved through complex numbers, complex vector spaces, concrete examples in functions and column vectors, the notion of dimension, the shift to abstract kets, the dual space, inner products, orthonormal bases, and the extraction of expansion coefficients.

Only after that apparatus was in place did the lecture turn to physics. Classical states form a set; quantum states form a vector space. Distinguishable configurations correspond to orthogonal vectors, superpositions are new states, normalization encodes total probability one, and squared amplitudes give measurement probabilities in a basis of distinguishable states. Finally, the lecture opened the next mathematical subject, operators, just far enough to show the direction of travel. The formalism is still abstract here, but the chapter ends with the same forward pressure as the lecture itself: we are now very close to the point where these rules begin to make concrete predictions.